EXPERIMENT NO 5

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PRN:-B25ET1201 Div:- SY3 Batch:- C3

INPUT CODE

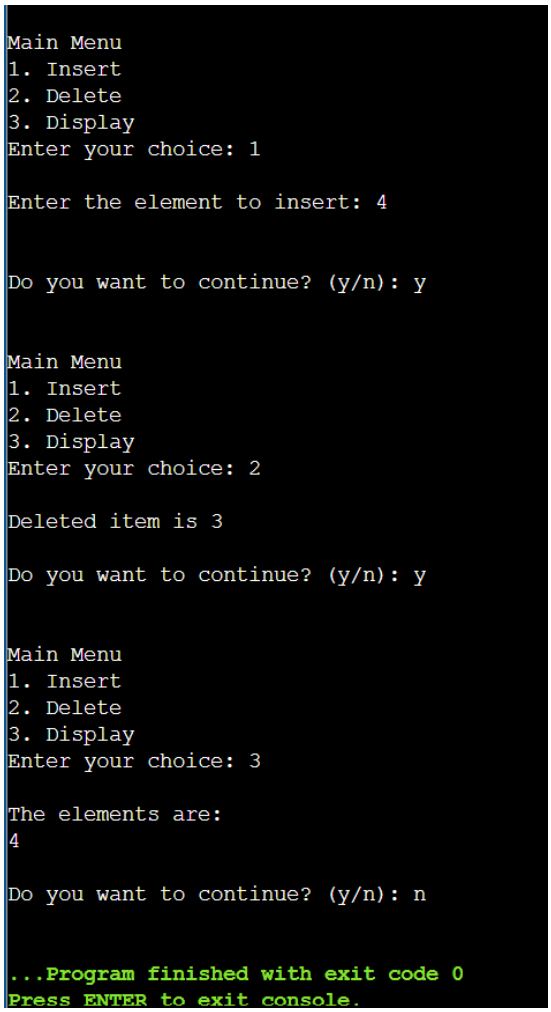
```
#include <stdio.h>
#define SIZE 5
struct Queue {
    int que[SIZE];
    int front, rear;
} Q;
int Qfull()
{
    if (Q.rear >= SIZE - 1)
        return 1;
    else
        return 0;
}
int Qempty()
{
    if (Q.front == -1 || Q.front > Q.rear)
        return 1;
    else
        return 0;
}
void insert(int item)
{
    if (Q.front == -1)
        Q.front = 0;
    Q.rear++;
    Q.que[Q.rear] = item;
}
int dequeue()
{
    int item;
    item = Q.que[Q.front];
    Q.front++;
    printf("\nDeleted item is %d", item);
```

```
    return item;
}
void display()
{
    int i;
    printf("\nThe elements are:\n");


    for (i = Q.front; i <= Q.rear; i++)
    {
        printf("%d ", Q.que[i]);
    }
}
int main()
{
    int choice, item;
    char ans;
    Q.front = -1;
    Q.rear = -1;
    do
    {
        printf("\n\nMain Menu");
        printf("\n1. Insert");
        printf("\n2. Delete");
        printf("\n3. Display");
        printf("\nEnter your choice: ");
        scanf("%d", &choice);
        switch (choice)
        {
            case 1:
                if (Qfull())
                {
                    printf("\nQueue is full");
                }
                else
                {
                    printf("\nEnter the element to insert: ");
                    scanf("%d", &item);
                    insert(item);
                }
                break;
            case 2:
                if (Qempty())
                {
```

```
                printf("\nQueue is empty");
            }
            else
            {
                dequeue();
            }
            break;
        case 3:
            if (Qempty())
            {
                printf("\nQueue is empty");
            }
            else
            {
                display();
            }
            break;
        default:
            printf("\nInvalid choice");
    }
    printf("\n\nDo you want to continue? (y/n): ");
    scanf(" %c", &ans);
  } while (ans == 'y' || ans == 'Y');
return 0;
}
```

OUTPUT


Main Menu
1. Insert
2. Delete
3. Display
Enter your choice: 1
Enter the element to insert: 4
Do you want to continue? (y/n): y
Main Menu
1. Insert
2. Delete
3. Display
Enter your choice: 2
Deleted item is 3
Do you want to continue? (y/n): y
Main Menu
1. Insert
2. Delete
3. Display
Enter your choice: 3
The elements are:
4
Do you want to continue? (y/n): n
...Program finished with exit code 0
Press ENTER to exit console.